

Palindromicity of antichain polynomials of three-dimensional boxes

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Abstract

For positive integers a, b, c , let

$$N_{a,b,c}(x) = \sum_A x^{|A|}$$

be the antichain generating polynomial of the product poset $[a] \times [b] \times [c]$. We prove that, after sorting $a \leq b \leq c$,

$$N_{a,b,c}(x) \text{ is palindromic} \iff (a, b, c) = (1, r, r) \text{ or } (2, r, r+1).$$

The new part is the exclusion of all boxes with $a, b, c \geq 3$. We use the last-passage representation of antichains by nonnegative matrices. In the monic parameter range

$$c = a + b - 1 - 2d,$$

we classify all binary supports one below maximum and all possible weight increments. For $d \geq 1$ this gives the exact coefficient

$$[x^{w-1}]N_{a,b,c}(x) = ab + 2 + (a-d)(a-d-1) + (b-d)(b-d-1),$$

where $w = ab - d(d+1)$. This coefficient is strictly smaller than $[x]N_{a,b,c}(x) = abc$. The remaining case $d = 0$ has an explicit coefficient strictly larger than abc . Thus the first and next-to-leading coefficients exclude palindromicity in every genuine higher-layer case. Combined with the known two-layer classification, this proves the rectangular minuscule branch of a conjecture of Ding and Dong.

1 Introduction

For a finite poset P , its antichain generating polynomial is

$$N_P(x) = \sum_{A \in \mathcal{A}(P)} x^{|A|},$$

where $\mathcal{A}(P)$ denotes the set of antichains of P . Ding and Dong studied palindromicity, real-rootedness, and gamma-positivity of these polynomials for minuscule posets [1]. Their Conjecture B states that, for a connected minuscule poset P , palindromicity of $N_{[k] \times P}(x)$ implies strict gamma-positivity.

The rectangular minuscule family takes

$$P = [b] \times [c],$$

so the relevant polynomial is that of the three-dimensional box $[a] \times [b] \times [c]$. The one-layer case is elementary. The two-layer case was recently classified together with a proof of real-rootedness and strict gamma-positivity [3]. The purpose of this paper is to settle every remaining number of layers.

Because the three factors are symmetric, we always sort

$$a \leq b \leq c.$$

Our main result is the following.

Theorem 1.1. *For positive integers $a \leq b \leq c$, the polynomial $N_{a,b,c}(x)$ is palindromic if and only if*

$$(a, b, c) = (1, r, r) \quad \text{or} \quad (a, b, c) = (2, r, r + 1).$$

The genuinely new assertion is:

Theorem 1.2. *If $3 \leq a \leq b \leq c$, then $N_{a,b,c}(x)$ is not palindromic.*

The proof needs only two coefficients, but obtaining the second one requires a structural classification. Amanov and Yeliussizov [2] give a last-passage correspondence between plane partitions and nonnegative matrices. Under this correspondence, the size of an antichain becomes the number of positive matrix entries. A near-maximum antichain is therefore a near-maximum support subject to a last-passage constraint.

The complement of the maximum support consists of two triangular end regions. Removing points from these triangles creates path deficits. We quantify the minimum cost of compensating those deficits inside the central band. A frontier recurrence first forces all removals to form one interval on the outermost diagonal of one end. A minimum-separator theorem then shows that the only compensation is the adjacent shadow, apart from two full cross-end exchanges. This yields exactly $ab + 2$ binary near-maximum supports. A separate dominator argument classifies all supports that can carry one additional unit of weight.

Section 2 recalls the last-passage model and the monic parameter reduction. Sections 3–5 prove the path-compensation classification. Section 6 classifies weight increments and obtains the next-to-leading coefficient for $d \geq 1$. Section 7 treats $d = 0$. Section 8 proves the complete classification.

2 Last passage and the monic parameter range

Write

$$Q = [0, a - 1] \times [0, b - 1].$$

For a nonnegative integer matrix $M = (M_{ij})_{(i,j) \in Q}$, let $L(M)$ be the maximum of

$$\sum_{(i,j) \in P} M_{ij}$$

over all northeast lattice paths P from $(0, 0)$ to $(a - 1, b - 1)$.

Amanov and Yeliussizov [2, Theorem 3.1, equation (3), Corollary 3.2(i), Lemma 3.4, and Lemma 4.2] give a bijection under which

$$N_{a,b,c}(x) = \sum_{\substack{M \in \mathbb{Z}_{\geq 0}^{a \times b} \\ L(M) \leq c}} x^{|\text{supp } M|}. \quad (2.1)$$

Indeed, their inverse map satisfies

$$M_{ij} > 0 \iff (i, j, \pi_{ij}) \text{ is a top corner of the corresponding plane partition.}$$

The top corners are the maximal elements of the corresponding order ideal, and hence form its antichain of maximal elements. This proves (2.1).

The constant term of $N_{a,b,c}(x)$ is 1. Consequently, a palindromic polynomial must be monic. Ding and Dong [1, Lemma 2.3(a)] classify the monic parameters. In the sorted range $a \leq b \leq c$, their condition is equivalent to

$$c = a + b - 1 - 2d, \quad 0 \leq d \leq \left\lfloor \frac{a-1}{2} \right\rfloor. \quad (2.2)$$

For these parameters, put

$$H = a + b - 2$$

and

$$B_d = \{(i, j) \in Q : d \leq i + j \leq H - d\}. \quad (2.3)$$

The unique maximum support is B_d , and its size is

$$w = |B_d| = ab - d(d + 1). \quad (2.4)$$

Thus non-monic parameters are already excluded. We shall compare $[x]N_{a,b,c}(x)$ and $[x^{w-1}]N_{a,b,c}(x)$ throughout the monic range. The first coefficient is immediate:

$$[x]N_{a,b,c}(x) = abc. \quad (2.5)$$

3 Path compensation

Assume in this section that

$$1 \leq d \leq \left\lfloor \frac{a-1}{2} \right\rfloor.$$

Let

$$T_0 = Q \setminus B_d.$$

Every full northeast path contains exactly $2d$ points of T_0 . Let S be a binary support of size $w - 1$, and write its complement uniquely as

$$Q \setminus S = (T_0 \setminus R) \cup A, \quad (3.1)$$

where

$$R \subseteq T_0, \quad A \subseteq B_d, \quad |A| = |R| + 1. \quad (3.2)$$

The binary matrix $\mathbf{1}_S$ satisfies $L(\mathbf{1}_S) \leq c$ if and only if

$$|P \cap A| \geq |P \cap R| \quad \text{for every full northeast path } P. \quad (3.3)$$

We call (R, A) a path-compensation pair.

For $0 \leq r \leq H$, define the rank diagonal

$$D_r = \{(i, j) \in Q : i + j = r\}$$

and, for $0 \leq r \leq d$, enumerate it as

$$D_r = \{x_{r,0}, \dots, x_{r,r}\}.$$

Let

$$R^- = R \cap (D_0 \cup \dots \cup D_{d-1}).$$

For $x_{r,k}$ define

$$v_r(k) = \max_P |P \cap R^-|, \quad (3.4)$$

where P ranges over paths from $(0, 0)$ to $x_{r,k}$. Then

$$v_r(k) = \mathbf{1}_{x_{r,k} \in R^-} + \max\{v_{r-1}(k-1), v_{r-1}(k)\}, \quad (3.5)$$

with nonexistent predecessors omitted. Define

$$G^-(R^-) = \sum_{k=0}^d v_d(k). \quad (3.6)$$

Lemma 3.1. *If r_0 is the lowest nonempty rank of R^- , then*

$$G^-(R^-) \geq |R^-| + (d - r_0). \quad (3.7)$$

Moreover,

$$G^-(R^-) = |R^-| + 1 \quad (3.8)$$

if and only if R^- is a nonempty consecutive interval in D_{d-1} .

Proof. For a nonnegative vector $u = (u_0, \dots, u_m)$, set

$$E(u) = (u_0, \max(u_0, u_1), \dots, \max(u_{m-1}, u_m), u_m).$$

Then

$$\sum E(u) - \sum u = u_m + \sum_{j=0}^{m-1} (u_j - u_{j+1})_+. \quad (3.9)$$

Starting at the lowest occupied rank and applying (3.5)–(3.9) at every subsequent rank proves (3.7). If $r_0 < d - 1$, at least two strict frontier expansions occur. If the lowest occupied rank has more than one consecutive component, the first expansion already creates at least two new frontier units. Any additional removal on a higher outer rank creates another unit in (3.5). Hence (3.8) forces a single interval in D_{d-1} . Conversely, an interval of length ℓ in D_{d-1} produces an interval of length $\ell + 1$ on D_d , so equality holds. $\square$

Rotating the rectangle through 180 degrees gives the analogous high-end quantities R^+ and $G^+(R^+)$.

Lemma 3.2. *Every path-compensation pair satisfies*

$$|A| \geq G^-(R^-) + G^+(R^+). \quad (3.10)$$

Consequently, R^- and R^+ cannot both be nonempty.

Proof. Write

$$y_p = (p, d - p), \quad z_p = (a - 1 - d + p, b - 1 - p), \quad 0 \leq p \leq d.$$

The inequality $2d \leq a - 1 \leq b - 1$ gives $y_p \leq z_p$. Use one fixed step word between each pair (y_p, z_p) . Its $d + 1$ translates form pairwise vertex-disjoint central channels.

At y_p , attach a low-end prefix attaining $v_d(p)$; at z_p , attach a high-end suffix attaining the corresponding rotated maximum. Applying (3.3) to these $d + 1$ full paths and summing, every point of $A \subseteq B_d$ is counted at most once, while the right side is exactly $G^-(R^-) + G^+(R^+)$. This proves (3.10).

If both ends contain removals, Lemma 3.1 and its rotation give

$$|A| \geq |R^-| + |R^+| + 2 = |R| + 2,$$

contrary to (3.2). $\square$

4 Minimum separators in the central band

We next classify the minimum vertex separators used to compensate a single end interval.

Lemma 4.1. *Let $U \subseteq D_{d+1}$ with $|U| = q \leq d + 1$. There are q pairwise vertex-disjoint northeast paths joining the points of U to q distinct points of D_{H-d} .*

Proof. Proceed one rank at a time. Before the final rank, every intermediate diagonal in the central band has at least $d+2$ points. For a proper subset Y of any such diagonal, its complete upper shadow has size at least $|Y|$. The only way a shadow can lose one point is for Y to fill an entire diagonal in the shrinking part of the rectangle. This cannot happen because there are only $q \leq d + 1$ active points.

Thus the bipartite graph between the current points and their successors on the next diagonal satisfies Hall's condition. Choose a matching at each rank. Composing the matchings gives the required disjoint paths. $\square$

Corollary 4.2. *Let $\emptyset \neq X \subseteq D_d$ with $|X| = k \leq d$. Any vertex set disjoint from X that meets every path from X to D_{H-d} has at least $k + 1$ points.*

Proof. The complete upper shadow of X satisfies

$$|\partial^+ X| = |X| + \#\{\text{consecutive components of } X\} \geq k + 1.$$

Choose $k + 1$ shadow points and apply Lemma 4.1. Extend the resulting paths down by one edge to X . They can share starting points in X , but outside X they are pairwise vertex-disjoint. A separator disjoint from X must therefore use at least one distinct point on each path. $\square$

Lemma 4.3. *Let $J \subseteq D_d$ be a nonempty consecutive interval of size m , and let $A \subseteq B_d$ meet every path from J to D_{H-d} . If $|A| = m$, then*

$$A = J, \tag{4.1}$$

except that when $J = D_d$ one may also have

$$A = D_{H-d}. \tag{4.2}$$

Proof. Put $X = J \setminus A$. If X is nonempty and $|X| = k \leq d$, then $A \cap J$ uses $m - k$ points, while Corollary 4.2 forces at least $k + 1$ further points. This exceeds the budget m . Therefore, if $m < d + 1$, one must have $A = J$.

Now let $m = d + 1$, so $J = D_d$. The preceding argument shows that A either equals D_d or avoids D_d completely. If A uses a nonempty proper subset of D_{H-d} , apply the rotated Corollary 4.2 to $D_{H-d} \setminus A$; the same budget contradiction results. Thus A is the whole upper boundary or avoids both boundaries.

In the latter case, apply the rank-by-rank Hall argument to the full D_{d+1} and D_{H-d-1} . It gives $d + 2$ internally disjoint paths. After extending them to the two boundary diagonals, they can share vertices only on boundaries that A avoids. Hence an internal separator needs at least $d + 2$ points, again a contradiction. $\square$

5 Binary supports one below maximum

Theorem 5.1. *Suppose $d \geq 1$. The binary supports $S \subseteq Q$ satisfying*

$$|S| = w - 1, \quad L(\mathbf{1}_S) \leq c$$

are exactly the following:

1. $S = B_d \setminus \{z\}$ for $z \in B_d$;
2. at either end, let I be a nonempty consecutive interval in the outer diagonal of T_0 , and let $J(I)$ be its complete shadow on the adjacent inner diagonal; then

$$S = Q \setminus ((T_0 \setminus I) \cup J(I));$$

3. the two cross-end full-diagonal exchanges

$$S = Q \setminus ((T_0 \setminus D_{d-1}) \cup D_{H-d})$$

and

$$S = Q \setminus ((T_0 \setminus D_{H-d+1}) \cup D_d).$$

Their total number is

$$ab + 2. \tag{5.1}$$

Proof. Use the complement representation (3.1). If $R = \emptyset$, then $|A| = 1$, giving the first family.

Assume $R \neq \emptyset$. Lemma 3.2 puts all removals at one end. Lemma 3.1 and $|A| = |R| + 1$ force R to be a single consecutive interval I in the outermost diagonal D_{d-1} , with no other removals.

Let

$$J = \partial^+ I \subseteq D_d.$$

Every path from J to D_{H-d} can be extended downward through a point of I and then to the source. Condition (3.3) therefore says that A is a separator for all such paths. Moreover,

$$|A| = |I| + 1 = |J|.$$

Lemma 4.3 gives $A = J$, except that when $I = D_{d-1}$ it also permits $A = D_{H-d}$. These are the same-end exchanges and one cross-end exchange. Rotation gives the high-end versions and the other cross-end exchange. Every listed pair directly satisfies (3.3).

There are w supports in the first family. At each end, the number of nonempty intervals in a diagonal of length d is

$$\sum_{\ell=1}^d (d - \ell + 1) = \frac{d(d+1)}{2}.$$

Adding both ends and the two cross-end exchanges gives

$$w + d(d + 1) + 2 = ab + 2.$$

This proves (5.1). □

6 Weight increments and the coefficient for $d \geq 1$

A matrix counted by $[x^{w-1}]N_{a,b,c}(x)$ may have entries larger than one. We now classify all such possibilities.

First consider

$$S = B_d \setminus \{z\}.$$

Increasing the entry at $u \in S$ from 1 to 2 is legal if and only if every saturated central-band path through u also passes through z . Equivalently, z dominates every source-to- u path or every u -to-sink path in the directed grid induced by B_d .

Lemma 6.1. *The legal ordered pairs (z, u) are exactly those on one of the four exposed boundary chains, with z separating the corresponding boundary endpoint from u . Their number is*

$$(a - d)(a - d - 1) + (b - d)(b - d - 1). \tag{6.1}$$

No matrix with two incremented points or an increment of at least two is legal.

Proof. If $u \in D_d$, there is no central-band point of smaller rank that can dominate it. If $\text{rank}(u) > d$ and u is not on a source-side exposed chain, two adjacent points of D_d reach u along two paths that meet only at u . Hence no $z \neq u$ lies on every source-to- u path. On an exposed chain, the path from its unique reachable boundary endpoint to u is forced, and its intermediate points are precisely the dominators. Rotation gives the sink-side statement, including the top boundary rank.

The four chain lengths are

$$a - d, \quad a - d, \quad b - d, \quad b - d.$$

Counting separated ordered pairs gives

$$2 \binom{a-d}{2} + 2 \binom{b-d}{2},$$

which is (6.1).

For fixed z , every legal u lies on one exposed chain. Any two such points occur together on a path that, after deleting z , has only $c - 1$ basic support points. Two unit increments, or two units at one point, raise its weight to $c + 1$. Thus no further increment is possible. □

Lemma 6.2. *Every point of every exchange support in Theorem 5.1 lies on a path with exactly c support points. Hence exchange supports admit no positive weight increment.*

Proof. Consider a low-end interval exchange. Let

$$I \subseteq D_{d-1}, \quad J = \partial^+ I \subseteq D_d.$$

If $u \in I$, extend a prefix through u to J and then to the sink; the path meets both I and J once. If $u \in D_d \setminus J$, enter u from a predecessor in $D_{d-1} \setminus I$; the path avoids both sets.

Finally, suppose $\text{rank}(u) > d$. The points of D_d that can reach u form an interval. If this interval meets J , enter through an intersection point and one of its predecessors in I . If it does not meet J , enter through a point outside J and a predecessor outside I . The resulting path still passes through u and meets I, J equally.

Thus every support point lies on a path satisfying

$$|P \cap I| = |P \cap J|.$$

Such a path has exactly c support points after the exchange. The high-end case follows by rotation. For a cross-end full exchange, every full path meets the removed and added diagonals once, so every path is already saturated. $\square$

Combining Theorem 5.1 and Lemmas 6.1–6.2 yields the exact coefficient.

Theorem 6.3. *For*

$$3 \leq a \leq b, \quad 1 \leq d \leq \left\lfloor \frac{a-1}{2} \right\rfloor,$$

put

$$c = a + b - 1 - 2d, \quad w = ab - d(d+1).$$

Then

$$\boxed{[x^{w-1}]N_{a,b,c}(x) = ab + 2 + (a-d)(a-d-1) + (b-d)(b-d-1).} \quad (6.2)$$

Moreover,

$$[x^{w-1}]N_{a,b,c}(x) < abc. \quad (6.3)$$

Proof. Theorem 5.1 counts $ab + 2$ binary supports. Lemma 6.1 contributes the two families of single increments in (6.1), and Lemma 6.2 excludes increments on every other support. This proves (6.2).

Expanding the difference gives

$$\begin{aligned} abc - [x^{w-1}]N &= (a-1)b^2 + (a-1)(a-1-2d)b \\ &\quad - (a-d)^2 - d^2 + a - 2d - 2. \end{aligned} \quad (6.4)$$

The second term is nonnegative. Since

$$(a - d)^2 + d^2 \leq a^2, \quad 2d \leq a - 1, \quad b \geq a,$$

we obtain

$$abc - [x^{w-1}]N \geq (a - 2)a^2 - 1 > 0.$$

This proves (6.3). □

7 The zero-depth case

It remains to treat the monic parameter

$$d = 0, \quad c = a + b - 1, \quad w = ab.$$

A support of size $ab - 1$ is $Q \setminus \{z\}$. Put weight 1 on this support and denote all additional weights by E . For every full path P , the constraint is

$$\sum_{u \in P} E_u \leq \mathbf{1}_{z \in P}. \tag{7.1}$$

If z is the source or sink, every path contains z . The nonzero support of E may then be any nonempty antichain other than $\{z\}$. The number of antichains in an a by b rectangle is

$$\binom{a+b}{a}.$$

Thus each endpoint contributes

$$\binom{a+b}{a} - 2$$

additional matrices beyond the binary matrix.

For non-endpoint z , an increment is possible only on the same exposed outer boundary, with z separating the endpoint from the incremented point. As in Lemma 6.1, no two increments and no larger increment are possible. The four boundary contributions total

$$2\binom{a-1}{2} + 2\binom{b-1}{2}.$$

Including the ab binary supports gives:

Proposition 7.1. *For $3 \leq a \leq b$,*

$$\begin{aligned} [x^{ab-1}]N_{a,b,a+b-1}(x) &= ab + 2 \left(\binom{a+b}{a} - 2 \right) \\ &\quad + (a-1)(a-2) + (b-1)(b-2) \\ &= 2 \binom{a+b}{a} + a^2 + b^2 + ab - 3a - 3b. \end{aligned} \tag{7.2}$$

This coefficient satisfies

$$[x^{ab-1}]N_{a,b,a+b-1}(x) > ab(a+b-1). \tag{7.3}$$

Proof. Only (7.3) remains. Its left side minus right side is

$$2 \binom{a+b}{a} - (a+b)(ab-a-b+3). \tag{7.4}$$

Set

$$R_{a,b} = \frac{2 \binom{a+b}{a}}{(a+b)((a-1)(b-1)+2)}.$$

For fixed $a \geq 3$, comparison of consecutive ratios gives

$$\begin{aligned} \frac{\binom{a+b+1}{a}}{\binom{a+b}{a}} - \frac{(a+b+1)((a-1)b+2)}{(a+b)((a-1)(b-1)+2)} \\ = \frac{(a-2)(a-1)(b-1)(a+b+1)}{(a+b)(b+1)(ab-a-b+3)} > 0. \end{aligned}$$

Thus $R_{a,b}$ increases with b . On the diagonal, let

$$L_a = 2 \binom{2a}{a}, \quad U_a = 2a((a-1)^2 + 2).$$

The base case is $L_3 = 40 > 36 = U_3$, and

$$\frac{L_{a+1}}{L_a} - \frac{U_{a+1}}{U_a} = \frac{(a-1)^2(3a^2-2a-2)}{a(a+1)(a^2-2a+3)} > 0.$$

Hence $R_{a,b} > 1$ for all $3 \leq a \leq b$, which is precisely (7.3). $\square$

8 Proof of the classification

Proof of Theorem 1.2. If $N_{a,b,c}(x)$ is not monic, it cannot be palindromic. In the monic range, write $c = a + b - 1 - 2d$ as in (2.2).

If $d = 0$, Proposition 7.1 gives

$$[x^{w-1}]N_{a,b,c}(x) > [x]N_{a,b,c}(x).$$

If $d \geq 1$, Theorem 6.3 gives the strict reverse inequality. In either case, the coefficients paired by palindromicity are unequal. Therefore $N_{a,b,c}(x)$ is not palindromic. $\square$

Proof of Theorem 1.1. When $a = 1$, the number of k -element antichains in $[b] \times [c]$ is

$$\binom{b}{k} \binom{c}{k}.$$

After sorting $b \leq c$, its top coefficient is $\binom{c}{b}$, so the polynomial is monic, and hence possibly palindromic, only when $b = c$. For $b = c = r$, the coefficients are $\binom{r}{k}^2$ and are palindromic.

When $a = 2$, the complete classification in [3] states that the polynomial is palindromic exactly when $c = b + 1$. When $a \geq 3$, Theorem 1.2 excludes every parameter. This proves the stated list. $\square$

Corollary 8.1. *Ding–Dong Conjecture B holds for the complete rectangular minuscule family.*

Proof. Theorem 1.1 lists all palindromic cases. For the one-layer square, Ding and Dong [1, Proposition 2.4] give the gamma expansion

$$\sum_{k=0}^r \binom{r}{k}^2 x^k = \sum_{j=0}^{\lfloor r/2 \rfloor} \frac{r!}{j! j! (r-2j)!} x^j (1+x)^{r-2j}.$$

Every allowed gamma coefficient is strictly positive. Reference [3] proves strict gamma-positivity for every palindromic two-layer case. There are no palindromic higher-layer cases. $\square$

9 Scope and verification

The proof is an all-parameter argument. Finite computation was used only for calibration and adversarial testing:

1. the Ding–Dong transfer recurrence reproduces their published example and scans small monic parameters;
2. direct complement enumeration verifies the binary support count and every possible single-cell unit increment for several $d = 1, 2$ cases, and retains a $d = 3$ negative control that disproved an earlier incomplete formula;
3. an independent row-frontier dynamic program verifies the binary support count in four $d = 3$ cases and obtains the value 75 for $(7, 7, 7)$ under the entry bound $0, 1, 2$.

These checks do not replace Lemmas 3.1–6.2. In particular, solver timeouts or finite absence of counterexamples are not used as existence or nonexistence proofs. Entries at least 3 and simultaneous increments are excluded by Lemmas 6.1–6.2, not by the finite scripts.

The result settles only the rectangular minuscule branch of Conjecture B. It does not claim a classification for the other connected minuscule posets.

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References

- [1] J. Ding and C.-P. Dong, *Antichain generating polynomials of posets*, arXiv:1905.06692v1 (2019). [arXiv:1905.06692](#).
- [2] A. Amanov and D. Yeliussizov, *MacMahon's statistics on higher-dimensional partitions*, Forum Math. Sigma 11 (2023), e63. [doi:10.1017/fms.2023.61](#).
- [3] Carptopus, *Real-rootedness, palindromicity, and gamma-positivity of antichain polynomials for $[2] \times [m] \times [n]$* , v0.2-beta (2026). [doi:10.5281/zenodo.22112708](#).